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CH.SC.U4CSE25275

Question_4:

Problem Description:

Dr. Malar was booking a tour package of IRCTC from Chennai to Delhi for his family.

Two of the relatives was interested in joining to this tour.

these two persons are studying engineering in computing technology. only one tickets are remaining in the IRCTC portal.

So, Dr. Malar decided to book one ticket for out of those persons also along with his family members.

she wants to identify the one person out of these persons. he decided to conduct a technical task to identify the right person to travel.

the task was that, implement two stack operations in an array

Can you help them to complete the task?

Constraints

0<n<5 only five elements has to be practiced for this operation

first element pushed into stack1

second element pushed into stack2, likewise elements pushed into alternative stacks vice versa.

Function Description

- Create a data structure *twoStacks* that represents two stacks.
- Implementation of *twoStacks* should use only one array, i.e., both stacks should use the same array for storing elements.
- Following functions must be supported by *twoStacks*.

  push1(int x) –> pushes x to first stack

  push2(int x) –> pushes x to second stack

  pop2() –> pops an element from second stack and return the popped element

  Implementation of *twoStack* should be space efficient.

Input Format

Single line represents only braces (both curly and square)

Output Format

If the given input balanced then print as Balanced (or) Not Balanced

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Logical Test Cases

Test Case 1

INPUT (STDIN)

```
1
2
3
4
5
```

EXPECTED OUTPUT

```
Popped element from stack1 is:5
Popped element from stack2 is:4
```

Test Case 2

INPUT (STDIN)

```
5
4
5
3
4
```

EXPECTED OUTPUT

```
Popped element from stack1 is:4
Popped element from stack2 is:3
```

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Code:

```
/*
Question_4:
Implement two stacks using one array.
Push alternate elements into stack1 and stack2,
then pop one element from each stack.
*/

#include <iostream>
using namespace std;

class twoStacks
{
    int arr[5];
    int top1, top2;

public:
    twoStacks()
    {
        top1 = -1;
        top2 = 5;
    }
    void push1(int x)
    {
        if(top1 + 1 < top2)
            arr[++top1] = x;
    }
    void push2(int x)
    {
        if(top1 + 1 < top2)
            arr[--top2] = x;
    }
    int pop1()
    {
        if(top1 == -1)
            return -1;
        return arr[top1--];
    }
    int pop2()
```

```
    {
        if(top2 == 5)
            return -1;
        return arr[top2++];
    }
};
int main()
{
    twoStacks ts;
    int x;
    for(int i = 1; i <= 5; i++)
    {
        cin >> x;
        if(i % 2 == 1)
            ts.push1(x);
        else
            ts.push2(x);
    }
    cout << "Popped element from stack1 is:" << ts.pop1() << endl;
    cout << "Popped element from stack2 is:" << ts.pop2() << endl;
    return 0;
}
```

Output:

Test Case_1:

```
1
2
3
4
5
Popped element from stack1 is:5
Popped element from stack2 is:4
```

Test Case_2:

```
5
4
5
3
4
Popped element from stack1 is:4
Popped element from stack2 is:3
PS C:\Users\Prabin Yadav\Documents\
```